

Genome-wide expression analysis of *HSP70* family genes in rice and identification of a cytosolic *HSP70* gene highly induced under heat stress

Ki-Hong Jung · Hyun-Jung Ghoo · Minh Xuan Nguyen · Sung-Ryul Kim · Gynheung An

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Abstract The heat shock protein 70 (HSP70) gene family plays a key role in protecting plant cells or tissues from thermal or oxidative stress. Although many studies have elucidated the molecular functions of individual family members, genome-wide analysis of this family is still limited, especially for crop species. Our objective was to integrate various meta-profiling data into the context of a phylogenetic tree, which would enable us to perform fine evaluation of functional dominance or redundancy within this family. Our data indicated that a loss-of-function mutant of a rice cytosolic *HSP70* gene (*OscHSP70-1*) did not show a clear defective phenotype in response to high temperature because of the existence of another gene family member that was closely clustered with *OscHSP70-1* and had similar expression patterns. Moreover, the second gene showed much stronger anatomical expression. We indirectly analyzed the function of *OscHSP70-1* by studying GUS activity under the control of the endogenous promoter. We also designed a probable interaction network mediated by *OscHSP70-1* and used co-expression analysis among its components to refine the network, suggesting more probable model to explain the function of *OscHSP70-1*.

Keywords Rice HSP70 family · Phylogenomics · Functional redundancy · Functional gene network analysis · Meta-profiling analysis

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K.-H. Jung (✉) · H.-J. Ghoo · M. X. Nguyen · S.-R. Kim · G. An
Department of Plant Molecular Systems Biotechnology and Crop
Biotech Institute, Kyung Hee University, Yongin 446-701, South
Korea
e-mail: khjung2010@khu.ac.kr

Abbreviations

ER	Endoplasmic reticulum
GEO	Gene expression omnibus
IDs	Identifiers
MS	Murashige and Skoog
PCD	Programmed cell death
RT-PCR	Reverse transcription polymerase chain reaction
TPR	Tetratricopeptide repeat
WT	Wild-type

Introduction

Global warming caused by elevated atmospheric temperatures is expected to trigger serious problems in crop supplement, similar to those already encountered by humans. Proteins belonging to a group of molecular chaperones, the 70-kDa heat shock proteins (HSP70s), are accumulated in response to a rapid rise in temperature (Sung et al. 2001). These chaperones are involved in various cellular processes, including correct folding of de novo synthesized proteins, protein transport across membranes, modulation of protein activity, regulation of protein degradation, and prevention of irreversible protein aggregation. Based on sub-cellular localization, the plant HSP70 gene family was originally divided into four subfamilies: cytosol, endoplasmic reticulum (ER), plastids, and mitochondria (Sung et al. 2001). In addition, *Nicotiana tabacum* contains a nuclear-localized HSP70, *NtHSP70-1*, which helps to prevent the fragmentation and degradation of nuclear DNA during heat stress (Cho and Choi 2009).

The biological functions of HSP70s have been investigated in several plant systems. Knockout mutants of *Arabidopsis thaliana* — *cpHSC70s*, *cpHSC70-1*, and *cpHSC70-2* — show

defective phenotypes in plant development and the thermotolerance of germinating seeds (Su and Li 2008). A deficiency in *AtHSP70-15* leads to severe growth retardation. Heat treatment of deficient plants results in a drastic increase in mortality, indicating that *AtHSP70-15* plays an essential role during normal growth and in the heat response (Jungkunz et al. 2011). Enhanced expression of *AtHSP70-1* confers stress tolerance whereas single and double knockout mutants of *HSP70-1* and *HSP70-2* do not show any growth defect (Jungkunz et al. 2011). Altered expression of the *A. thaliana* cytosolic/nuclear HSC70-1 molecular chaperone affects development and abiotic stress tolerance, and the interaction between *Arabidopsis* SGT1 and cytosolic/nuclear HSC70 chaperones modulates immune responses (Noel et al. 2007). Another gene from *A. thaliana*, *BIP*, acts in the fusion of polar nuclei during female gametophyte development (Maruyama et al. 2010), while the same gene in *N. tabacum* confers protection against water stress (Alvim et al. 2001). In rice, the *BIP1/OsBIP3* gene encodes HSP70 in the ER, which interferes with XA21-mediated immunity through its interaction with XA21 (Park et al. 2010). Overexpression and knock-down analyses of the *BIP1/OsBIP3* gene have demonstrated defects in seed development (Wakasa et al. 2011), and a gain-of-function study with a mitochondrial HSP70 showed suppression of programmed cell death (Qi et al. 2011). However, the biological functions of many HSP70s have not yet been elucidated, perhaps because of functional redundancy as demonstrated in a loss-of-function study of *Arabidopsis* cytosolic *HSP70* genes (Jungkunz et al. 2011).

Rice interaction viewer queries predicted a rice interactome consisting of 37,472 predicted and 430 confirmed rice protein interactions (http://bar.utoronto.ca/interactions/cgi-bin/rice_interactions_viewer.cgi). The fidelity of those predicted interactions can be enhanced by evaluating co-expression patterns, recorded phenotypes, sub-cellular locations, and biological processes among those interlogs (Jin et al. 2013).

Here, we present an initial phylogenomics analysis of the rice HSP70 family by integrating multi-omics high-throughput data into the context of a phylogenetic tree. We then carried out functional analysis of a cytosolic HSP70 (*OsctHSP70-1*) family member that was significantly stimulated under heat stress using mutant lines with T-DNA in this gene, but did not identify defective phenotypes due to functional redundancy with another cytosolic HSP70 that had similar expression patterns with respect to anatomy and stress conditions. Development of a predicted interaction network for *OsctHSP70-1* and a GUS reporter assay under control of the endogenous promoter provide alternative ways to identify gene function. Considering co-expression patterns among interacting pairs in response to heat stress further enhances the probability strength of the network.

Fig. 1 Phylogenomics analysis of rice and *Arabidopsis* Hsp70 proteins. **a** Phylogenetic tree developed from 32 rice and 18 *Arabidopsis* HSP70 proteins using MEGA version 5.0 software. **b** Integration of *Arabidopsis* ortholog locus IDs, gene names, and subcellular localization information into the context of the phylogenetic tree consisting of 32 Hsp70 protein sequences. Cluster I, green; Cluster II, blue; Cluster III, light green; Cluster IV, dark green; Cluster V, light brown; Cluster VI, pink. The numbers in the phylogenetic tree are bootstrap values tested with 500 replicates using MEGA version 5.0 software. The numbers below the phylogenetic tree indicate evolutionary distance calculated by phylogenetic analysis as described in Materials and methods

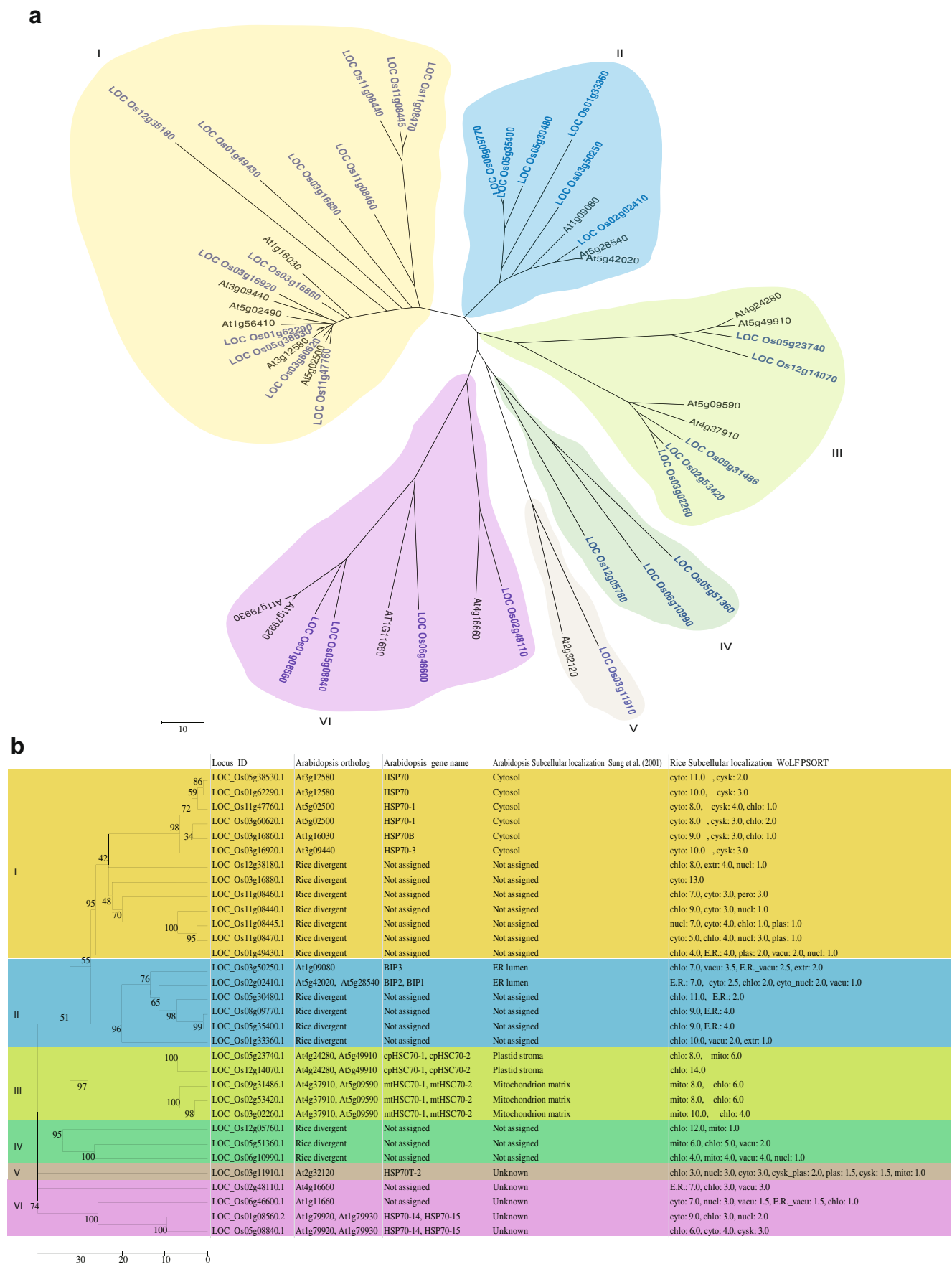
Materials and methods

Phylogenetic analysis of the rice HSP70 family and integration of public microarray data

To develop a phylogenetic tree, we used 32 HSP70 protein sequences identified from the Greenphyl phylogenomics database. The sequences were prepared in a FASTA formatted file (Table S1) and loaded into the ClustalX2 program for multiple alignments. The alignment file format (.aln) was converted to a mega file format using the file converter option installed in MEGA version 5.0 (Tamura et al. 2011). We then produced a phylogenetic tree for these proteins using MEGA version 5.0 under the following parameters: neighbor-joining tree method, complete deletion, and bootstrap with 500 replicates. After obtaining 18 *Arabidopsis* HSP70 protein sequences from the Greenphyl database, we developed a combined phylogenetic tree for the rice and *Arabidopsis* HSP70 protein families by the neighbor-joining method (Fig. 1). We then integrated anatomical expression data for 27 of the 32 genes extracted from the anatomical meta-profile-based 209 Agilent 44 K array (GSE21494) (Sato et al. 2011b) (Fig. S1, Table S2). Anatomical meta-profiling data using 983 Affymetrix arrays for the 32 *HSP70* genes are shown in Fig. S2. We also incorporated the log₂ fold-change data in response to drought (GSE6901), salt (GSE6901), cold (GSE6901), and late heat for 10 h at 42 °C (GSE14275) from the Affymetrix data, as well as data for light versus dark treatment in the young seedling stage from the NSF45K array, within the context of the phylogenetic tree (Jung et al. 2008). For early heat responses, we generated NSF45K array data to compare leaves after 0.5 or 1.0 h at 37 °C for exposure to high-temperature stress versus those examined at 0 h (control, 28 °C) (Jung et al. 2012). The log₂ fold-change data for heat treatment over control readings were used to produce the heat map depicted in Fig. 2.

Plant growth and exposure to various stress treatments

We germinated wild-type (WT) rice (*Oryza sativa* L. japonica genotype Dongjin) and B49 seeds (a promoter trap line of *OsctHSP70-1*; *Os05g38530*) in Murashige and Skoog (MS)



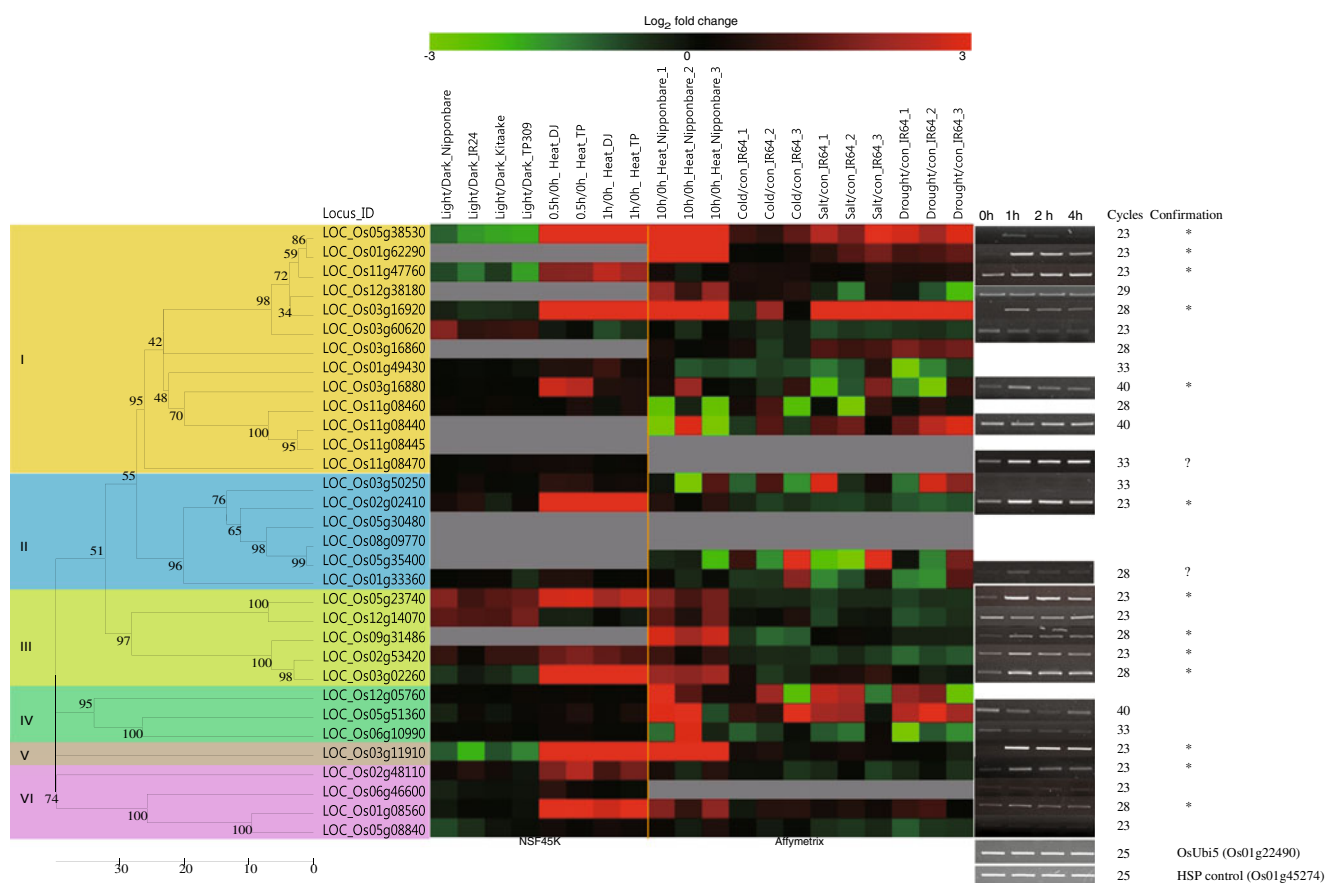


Fig. 2 Phylogenomics analysis of the rice Hsp70 family by integrating fold-change data in response to light, dark, heat, drought, salt, or cold. Log₂ fold-change data for NSF45K data were used to compare light versus dark and early heat stress versus control. Affymetrix array data were used to compare late heat stress versus control, cold versus control, salt versus control, and drought versus control. All data were integrated into the phylogenetic tree of the Hsp70 family. RT-PCR analyses were performed with leaf tissues collected after 0, 1, 2, or 4 h at 42 °C. *OsUbi5* (*Os01g22490*) and HSP control (*Os01g45274*)

were identified from the NSF45K array to compare early heat stress versus control and showed consistent expression. In stress conditions versus control, *red* indicates up-regulation; *green*, down-regulation; and *gray*, data not available. *Asterisks* indicate genes with significant up-regulation in response to heat stress based on microarray analysis and RT-PCR. A *question mark* indicates a gene with inconsistent expression results between microarray and RT-PCR. Data are representative of two independent RT-PCR analyses

medium containing 0.44 % MS basal salts, 3 % sucrose, 0.2 % Phytigel, and 0.55 mM *myo*-inositol. After incubation at 28 °C for 1 week under continuous light, seedlings were transferred to tap water for 2 days of adaptation before beginning the stress experiments.

For high-temperature treatment, the germinating seedlings were transferred to a beaker containing water pre-heated to 42 °C and incubated for up to 4 h in a thermo incubator (N-biotek, Bucheon-si, South Korea) set at 42 °C. To induce drought stress, we placed whole seedlings on a paper towel and exposed them to room temperature and humidity. For salt treatment, the seedlings were incubated for up to 4 h in a beaker containing a 200-mM NaCl solution. Cold treatment was applied by incubating seedlings for up to 4 h in a beaker containing water pre-chilled to 4 °C. Whole seedlings were collected from each stress treatment after 0, 1, 2, or 4 h of exposure for use in GUS assays. GUS staining of

root samples was performed as described by Jefferson et al. (1987) and the roots were photographed with an EOS 560 digital camera (Canon, Japan). For heat treatment, we also extracted RNA for gene expression analysis.

RT-PCR analysis

We isolated total RNA (50–100 µg) from leaves of 9-day-old seedlings after 0, 1, 2, or 4 h of exposure to high temperature (42 °C) using Trizol reagent and treated it with DNaseI for 15 min before purification with an RNeasy Midi Kit (Qiagen, USA). Quantities of total RNA and mRNA were determined by measuring *A*₂₆₀ and *A*₂₈₀ with a Nanodrop ND-1000. The level of protein contamination in the RNA was estimated based on the *A*₂₆₀ to *A*₂₈₀ ratio and only RNA samples with ratios of 2.0 to 2.2 were used for these experiments. cDNA was synthesized from 2 µg of total RNA using Moloney

murine leukemia virus reverse transcriptase (Promega, USA) in the appropriate reaction buffer. Polymerase chain reaction (PCR) was performed in a 30- μ l solution containing a 1- μ l aliquot of the cDNA reaction, 0.2 μ M gene-specific primers, 10 mM dNTPs, 1 unit of ExTaq DNA polymerase (Takara, Japan), and reaction buffer. The reaction included an initial 5-min denaturation at 94 °C, followed by 21 to 40 cycles of PCR (94 °C for 45 sec, 60 °C for 45 sec, and 72 °C for 1 min), and a final 10-min incubation at 72 °C. The balance of cDNA synthesis was evaluated by RT-PCR for rice *Ubiquitin 5* (*OsUbi5/Os01g22490*; forward primer 5'-GCACAAGCACAAGAAGGTGA-3' and reverse primer 5'-GCCTGCTGGTTGTAGACGTA-3'), rice *Ubiquitin 1* (*OsUbi1/Os03g13170*; forward primer 5'-TGAA GACCTGACTGGGAAG-3' and reverse primer 5'-CACG GTTCAACAACATCCAG-3'), and carbonyl anhydrase (HS control/*Os01g45274*; forward primer 5'-CCCTCCTCTC ACTCAAGGAT-3' and reverse primer 5'-ACGGGTAG GTCTTGAGGTTC-3'). We then analyzed expression patterns for 31 network genes that were up-regulated in response to heat stress. Primer sequences are listed in Table S3.

Interaction network analysis

To develop a functional gene network mediated by *OsctHSP70-1*, we used the rice interaction viewer and found 175 genes that interact with *OsctHSP70-1*. We refined the network by identifying 19 heat-inducible genes after integrating fold-changes in response to early heat stress. We then further refined those interactors by performing hierarchical clustering analysis with an anatomical meta profile for 20 heat-responsive genes, including *OsctHSP70-1*. Three genes had expression patterns similar to *OsctHSP70-1*. Moreover, two interactors had predicted locations in the cytoplasm that were similar to that of *OsctHSP70-1*. We then expanded the network by identifying proteins with five key interactors. This enlarged network was further refined by integrating the fold-change data for response to heat, and by selecting proteins connected to at least two proteins in the network. Finally, expression patterns for 31 of 34 genes refined in the expanded functional network were confirmed by RT-PCR.

Results

Phylogenetic analysis of the rice HSP70 family coupled with functional classification by subcellular localization

We identified 32 rice HSP70 family members from the Greenphyl phylogenomics database (Rouard et al. 2011). A recent study reported the same number of HSP70 superfamily members in rice (Sarkar et al. 2012). The general functions of

individual members of this family include protein folding and responses to heat stress. For phylogenetic analysis, we also identified 18 *Arabidopsis* HSP70 proteins from the Greenphyl phylogenomics database. Although it is debatable whether At1g11660, At1g79220, At1g79930, and At4g16660 are truly members of the HSP70 family, we included these in further analyses. Our phylogenetics analysis with 32 rice and 18 *Arabidopsis* HSP70 proteins revealed six subfamilies (Fig. 1a). We then developed a phylogenetic tree of 32 rice HSP70 proteins using this subfamily information. In addition, we integrated the *Arabidopsis* ortholog locus identifiers (IDs), the gene names, and the subcellular localizations reported by Sung et al. (2001) into the phylogenetic context (Fig. 1b). Cluster I has 13 members, and six of them with *Arabidopsis* orthologs localized in the cytoplasm, and seven of these form a cluster among rice HSP70s and are defined as a rice divergent group. Cluster II has six family members, two of which are predicted to be localized in the ER lumen based on *Arabidopsis* orthologs, while the others belong to a rice divergent group. Cluster III consists of five members with *Arabidopsis* orthologs, two in the plastid stroma and three in the mitochondrion matrix. Cluster IV has three members belonging to a rice divergent group. Cluster V, with a single member, and Cluster VI, with four members, lacked information on subcellular localization based on *Arabidopsis* orthologs. We used the WoLF PSORT tool (Horton et al. 2007) to predict these HSP70s and found that Os02g48110 contains an ER retention signal (HDEL) at the C terminus and is therefore predicted to localize to the ER. In addition, Os06g46600 and Os01g08560 are predicted to localize to the cytoplasm with the highest probability, and Os05g08840 to the chloroplast. However, the subcellular localization of Os03g11910 was not specified to a single cellular compartment. Therefore, based on results from our integration analysis of phylogenetic tree and subcellular localization prediction, we could speculate that the rice HSP70 proteins function in different compartments. Specifically, Cluster I is preferentially functional in the cytoplasm, Cluster II in the ER, and Cluster III in plastid stroma and the mitochondrion matrix.

Integrating meta-expression profiles in response to abiotic stresses reveals 13 heat stress-responsive HSP70 genes

Recently, Ye et al. (2012) analyzed the expression patterns of nine genes and Hu et al. (2009) presented transcriptome data for 26 genes in response to abiotic stresses. In our study, using Affymetrix gene chip data we integrated the differential expression patterns of rice genes in response to salt, drought, cold, or late heat into the phylogenetic tree context of 32 rice HSP70s (Hu et al. 2009; Jain et al. 2007). We also used the NSF45K array to confirm the data for differential expression patterns by comparing seedlings grown in the light versus dark (GSE8261), and those exposed to early

heat stress (0.5 or 1.0 h at 37 °C) versus the control (28 °C) (Fig. 2, Table S4) (Jung and An 2012; Jung et al. 2008, 2012).

In Cluster I, *Os05g38530* (*OscHSP70-1*), *Os01g62290*, *Os11g47760*, *Os12g38180*, *Os03g16920*, and *Os03g16880* showed a greater than 2-fold (i.e., significant) induction in response to heat stress. Of these, *Os03g16880* was up-regulated after 0.5 h of treatment whereas *Os11g47760* were up-regulated after 0.5 and 1.0 h, respectively. *Os05g38530* and *Os03g16920* were up-regulated by at least 10-fold after 0.5, 1.0, and 10.0 h, demonstrating their response to both early and late heat stress. *Os01g62290* and *Os12g38180* responded to 10 h of exposure at 42 °C, but data for their response to early heat stress were not available because they were not printed on the NSF45K array. However, because the expression patterns between *Os01g62290* and *Os05g38530* were very similar under other treatments, we expected that the former would be responsive to early heat.

Os05g38530, *Os01g62290*, and *Os03g16920* were also significantly up-regulated by salt and drought, indicating that they might have roles in various abiotic stress responses. *Os03g16880* was up-regulated only 2-fold upon drought and salt treatments, and its response to heat was unavailable. *Os05g38530*, *Os11g47760*, and *Os03g16920* were down-regulated by light whereas *Os01g49430* was down-regulated by salt and cold. *Os03g60620* was down-regulated by drought and salt but up-regulated by light. Based on those observations, we further divided these proteins into two subgroups: light-responsive (upper panel in Fig. 2) and light-unresponsive (lower panel in Fig. 2).

The rice *OscHSP70* genes were up-regulated in response to drought, salt, and heat, with *Os05g38530*, *Os01g62290*, and *Os03g16920* showing the greatest response. As expected from their anatomical expression patterns, *Os05g38530* and *Os01g62290* may have functional redundancy, suggesting that mutation of a single gene might not display obvious phenotypic defects.

In Cluster II, *BIP1/OsBIP3* (*Os02g02410*) showed strong up-regulation in response to early heat stress, but down-regulation following treatment with drought, salt, or cold. Based on anatomical expression patterns, we predicted that this gene has a dominant role in the early heat response. In Cluster III, *Os05g23740* and *Os12g14070* had light- and heat-inducible expression, but they differed in their response to early heat stress, with the former being up-regulated and the latter being down-regulated. Expression of *Os05g23740* was suppressed by drought, salt, and cold, while that of *Os12g14070* was inhibited by drought and salt. Their light- and heat-inducible expression patterns implied that these *OscpHSP70* genes function in light-dependent heat-stress responses in the leaf blade as well as during endosperm development. The responses of *Os05g23740* to cold, salt, and drought were antagonistic to those observed with light

and heat treatments (Fig. 2), indicating that such responses in the chloroplasts might be stimulated by the restriction of other abiotic stress responses. *Os03g02260* was down-regulated by light and up-regulated by heat like several *OscHSP70* genes (i.e., *Os05g38530*, *Os11g47760*, and *Os03g16920*). Both *Os02g53420* and *Os03g02260* showed early and late heat-inducible expression patterns. Likewise, *Os02g53420* was down-regulated by cold, drought, and salt while *Os09g31486* was down-regulated by cold. By comparison, *Os03g02260* did not respond to any stress other than heat. Although *Os09g31486* was responsive to late heat, its response to early heat could not be analyzed in the NSF45K array.

No genes in Cluster IV showed significant differential expression patterns in response to heat. Drought conditions caused *Os05g51360* to be up-regulated while *Os06g10990* was down-regulated. In Cluster V, *Os03g11910* was up-regulated by both early and late heat, but down-regulated by light. The antagonistic expression pattern for *Os03g11910* with regard to light and heat was very similar to that of *Os05g38530* and *Os03g16920*. In Cluster VI (*OsnHSP70* genes), *Os02g48110* and *Os01g08560* were up-regulated by early heat, and the latter was also up-regulated by late high temperature. *Os05g08840* was down-regulated by drought or salt treatment.

In total, we identified 13 *HSP70* genes with at least 2-fold up-regulation in response to either early or late heat stress. Most cellular compartments contained at least one heat-responsive *HSP70* gene, indicating that rice has differentiated *HSP70*s unique to each cellular compartment. These genes modulate diverse biological or cellular functions such as the thermotolerance response, protein folding, protein transport across membranes, regulation of protein degradation, and prevention of irreversible protein aggregation. However, we could not analyze the heat stress responses for nine genes on the NSF45K array and five on the Affymetrix array. Therefore, we still need to evaluate the expression patterns for those missing genes and also confirm the expression patterns of *HSP70* genes printed on the microarrays.

Because our primary interest was how *HSP70* genes mediate the response to high temperature, we used RT-PCR and qRT-PCR to evaluate the expression patterns of 32 such genes in 7-day-old seedlings following exposure to 42 °C for 0, 1, 2, or 4 h. As a stable internal control we used *OsUbi5* (*Os01g22490*) (Jain et al. 2006), and *Os01g45274* encoding carbonic anhydrase because public microarray data (GSE14275) demonstrated that the latter had stable expression in response to heat stress. Expression was detected for 24 out of those 32 *HSP70* genes, indicating that at least these 24 were functional in the early seedling stage. For the remaining eight genes, we identified bands for the primer sets with genomic DNA, but could not amplify them by RT-PCR even after more than 40 cycles (data not shown). Fourteen out of 24 genes

were significantly up-regulated by exposure to 42 °C based on results from our RT-PCR analyses (Fig. 2).

Thirteen of the genes had similar trends in expression, as shown in the microarray experiments (marked with asterisks in Fig. 2). *Os05g38530* (cytoplasm, *OscHSP70-1*), *Os01g62290* (cytoplasm), *Os03g16920* (cytoplasm), *Os03g16880* (cytoplasm), *BIP1/OsBIP3* (*Os02g02410*, ER), *Os05g23740* (chloroplast), *Os03g11910* (cytoplasm), *Os02g48110* (nucleus), and *Os01g08560* (nucleus) showed peak transcript levels at 1 or 2 h after heat stress was applied. As expected, *Os05g38530* and *Os01g62290* had similar expression patterns in response to early heat, suggesting functional redundancy between them. This redundancy may also include *Os03g16920* based on differential expression patterns in response to various abiotic stresses (Fig. 2). One cytosolic *HSP70* gene, *Os11g47760*, showed proportional increases in expression between heat inducibility and stress duration, with its expression peaking after 4 h of treatment (Fig. 2). Down-regulation of *Os03g60260* after 1, 2, and 4 h of stress was confirmed by RT-PCR. Along with previously known heat-inducible genes that were confirmed in various compartments, our RT-PCR analyses also identified new early heat-inducible *HSP70* genes by microarray analyses. For example, *Os11g08470* was up-regulated in response to 1, 2, and 4 h of stress. Our findings provided further evidence that major sub-cellular organelles in rice have machinery for protein folding or denaturation, mediated by the *HSP70* proteins specific to each organelle.

Functional analyses of an *OscHSP70-1* gene using a T-DNA gene-indexed line

We identified a T-DNA insertion line (Line B49) of the *OscHSP70-1* gene. Within Cluster I, *Os05g38530* was the most significantly induced in response to heat stress. Therefore, we assayed its response after exposing the homozygous and WT progenies to 42 °C but found no clear phenotypic differences between them. This indicated that the function of another *OscHSP70-1* gene, *Os01g62290*, in the same cluster might be sufficient to compensate for the defects caused by the T-DNA insertion in *Os05g38530*. To check this possibility, we examined the expression levels of *Os01g62290* in leaves from homozygous and WT progenies of the T-DNA insertion line in *Os05g38530*. Expression of *Os01g62290* was up-regulated in the homozygotes (KO in Fig. S3) but not in the WT progenies (WT-1 and WT-2 in Fig. S3) among the T2 generation with a T-DNA insertion in the *OscHSP70-1* gene. This implied that expression of *Os01g62290* in the mutant was stimulated to compensate for the defect of *Os05g38530*, and may have been sufficient to replace the role of the latter, which was expressed at a relatively low level in most anatomical samples with similar heat-responsiveness (Fig. S1). In addition, we examined the

expression patterns of *Os11g47760*, a gene very closely clustered with *Os01g62290* and *Os05g38530*. Expression by *Os11g47760* was not affected by the *Os05g38530* mutation, thereby indicating that its functional redundancy was restricted to *Os01g62290* and not expanded to *Os11g47760*.

Expression profiles of an *OscHSP70-1* gene in response to heat stress using the beta-glucuronidase (GUS) reporter gene

We previously identified 383 lines with GUS activity in their leaves, roots, flowers, and immature seeds (Jeon et al. 2000). One line, B49, has a T-DNA insertion in the first intron of *Os05g38530*, which contains two exons and one intron (Fig. 3a). Co-segregation of the insertion and GUS expression was verified in the T2 generation by genotyping analysis and GUS assay (Jung et al. 2003). We also performed RT-PCR on homozygous seedlings carrying the insertion to confirm the fusion transcript of the first exon of *Os05g38530* and *GUS* after a splicing event in which the donor was in the first intron of *Os05g38530* and the acceptor was positioned upstream of *GUS* (data not shown). Therefore, GUS activity could be used to represent the expression patterns of the *Os05g38530* gene under its native promoter.

Our phylogenomics data allowed us to identify genes that were significantly stimulated in response to heat, drought, or salt, but not to cold. Among these, the heat response was the most obvious. Therefore, we first exposed heterozygous seedlings of Line B49 to 37 °C for 0, 1, 2, or 4 h and examined the temporal expression patterns of GUS activity (Fig. 3b). We also monitored activity under the control of the *OscHSP70-1* promoter that occurred in the roots and leaves of heterozygous seedlings after 1 h of exposure. Activity was significantly increased after 2 h and was nearly saturated by 4 h. This demonstrated that *Os05g38530* was immediately activated by heat stress at the transcriptional level and that the protein was proportionally accumulated in the first 4 h of treatment. *GUS* expression patterns were confirmed by RT-PCR (Fig. 3b) using transverse sections prepared from GUS-stained roots and leaves of Line B49. Expression was more prominent in the phloem than in any other leaf tissues (Fig. 3c). Similarly, *GUS* was preferentially expressed in the protophloem of seminal roots and in the protophloem and endodermis of crown roots and their lateral roots (Fig. 3e, g). Expression was greater in the crown roots than in the seminal roots, and was much stronger in the protophloem of the elongation zone than in the maturation zone. The lateral roots of both seminal and crown roots showed strong GUS activity, implying that they might be sites of the primary response to heat stress.

We also examined GUS activity in Line B49 in response to drought, salt, and cold, but found that it was less obvious than the expression stimulated by heat stress (data not shown). Based on our analysis of GUS activity, we concluded that expression in rice crown roots is the primary mechanism for

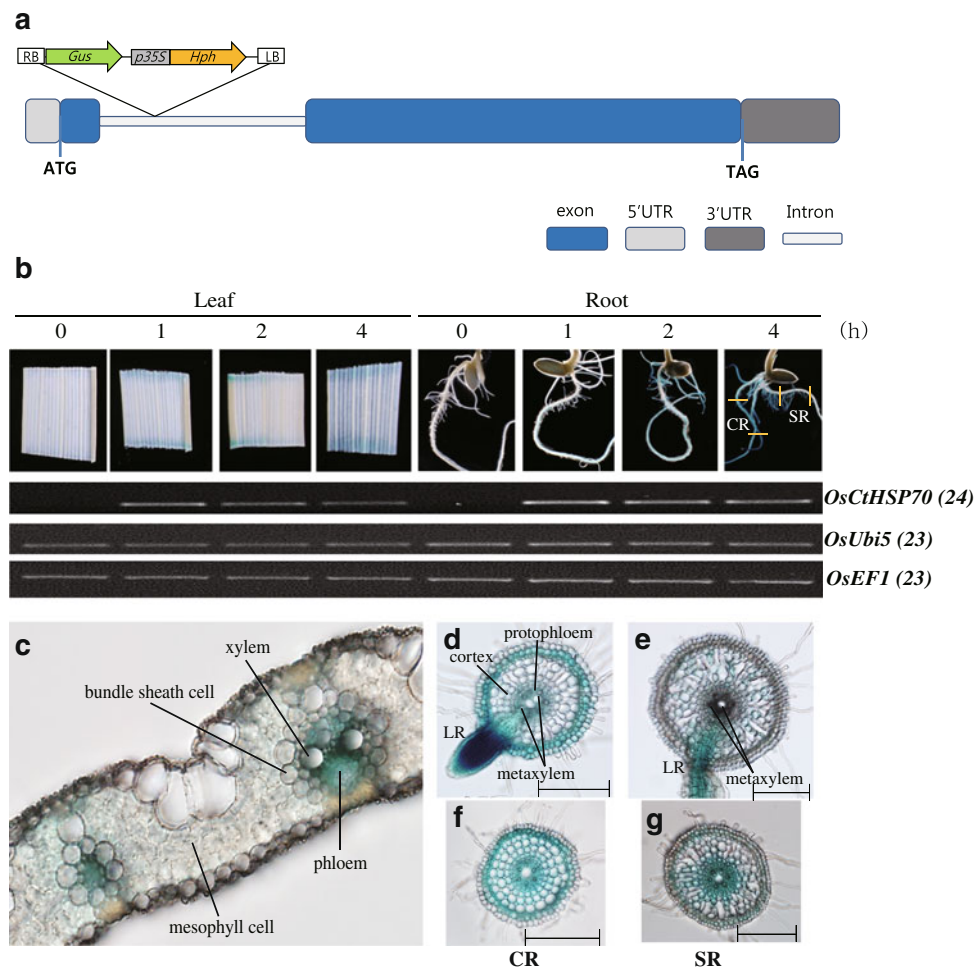


Fig. 3 GUS activity in promoter trap line (B49) for rice *OsctHsp70-1* after heat-stress treatment. **a** Schematic diagram of T-DNA insertion with the promoterless *GUS* reporter gene in the first intron of *OsctHsp70-1*. After a splicing event between the first intron donor and an acceptor upstream of the *GUS* coding sequence, a fusion transcript was generated between the first exon and the coding sequence. Blue box exon, light-gray box 5'UTR, dark-gray box 3'UTR, light-gray line intron, ATG start codon of *OsctHsp70-1*, TAG stop codon of *OsctHsp70-1*, RB right border of T-DNA, LB left border of T-DNA, *GUS* beta-glucuronidase, *Hph* hygromycin phosphotransferase, *p35S* cauliflower mosaic virus 35S promoter. **b** GUS activity under control of *OsctHsp70-1* promoter and expression patterns for *OsctHsp70-1* after

treatment at 37 °C. Homozygous progenies of Line B49 carrying a T-DNA insertion in *OsctHsp70-1* were grown on MS medium for 1 week. Whole seedlings were then exposed to 42 °C for up to 4 h. Stressed seedlings were sampled at time zero and after 1, 2, or 4 h of treatment. For RT-PCR, rice *Ubiquitin 5* (*OsUbi5*, *Os01g22490*) and *Elongation factor 1 alpha* (*OsEF1α*, *Os03g08010*) served as internal controls for cDNA levels in different samples. *GUS* expression patterns were monitored in 7-day-old leaf cross section (**c**), maturation zone (**d**) and elongation zone (**f**) of crown roots, and maturation zone (**e**) and elongation zone (**g**) of seminal roots from *GUS*-stained Line B49 after 4 h of heat treatment. LR lateral root. Scale bar = 100 μm. Green color indicates *GUS* activity

protection of plants against high temperatures mediated by *OsctHSP70-1*.

Development and refinement of a functional network mediated by *OsctHSP70-1*

We attempted to develop an interaction network originating from *OsctHSP70-1* (*Os05g38530*) using the rice interaction viewer tool. We also produced a putative interaction map of the HSP70 protein based on the identification of interlogs that have interaction partners in worm, fly, human, mouse, and *Escherichia coli*. Using this procedure, we found 175

proteins that interact with *OsctHSP70-1*, indicating that it performs diverse roles (Fig. 4). After integrating the fold-change data for responses to early heat stress into the node color in the network, we then refined the network by selecting 19 interactors with at least 2-fold up-regulation. From these, we further refined the network by analyzing co-expression patterns among those 19 genes and *OsctHSP70-1* using 983 Affymetrix data (Cao et al. 2012). In all, three genes had co-expression patterns with *OsctHSP70*: *Os05g48810*, encoding a dnaJ domain (HSP40)-containing protein; *Os05g44340*, encoding heat shock protein 101 (HSP101); and *Os03g53910*, encoding a tetratricopeptide repeat (TPR) domain-containing protein (Fig. 4). We also

identified two interactors (T-complex protein and Ras-related protein) with GO terms in the same cellular component category as *OscHSP70-1* (Fig. 4).

We used the rice interaction viewer to examine how the interaction network is related to *OscHSP70-1* and five key interactors. In all, 28 interactors were identified with *Os05g48810*, 2 with *Os05g44340*, 212 with *Os10g32550*, 28 with *Os01g42530*, and 19 with *Os03g53910* (Fig. 4). This was in addition to the 175 previously found (above) for *OscHSP70-1*. Thus, we could predict 454 interactions among 421 genes. We again refined the network by selecting 14 genes that were up-regulated at least 2-fold in response to heat stress, as well as 24 interactors that were connected at least twice with *OscHSP70-1* or five key interactors. This refined network comprised 34 up-regulated heat-responsive genes (including *OscHSP70-1*) and 24 genes with multiple connections to *OscHSP70-1* or those five key interactors. Expression patterns were validated by RT-PCR for 31 of those 34 up-regulated genes, such that the network was

further refined to 31 components by co-expression analysis and to 24 components by multiple interactions. This network, with annotation information in the nodes, is depicted in Fig. 5; the locus ID version is presented in Fig. S4.

Discussion

Expression analysis using a reporter system to address the limitations due to functional redundancy

The advantage of using the GUS assay under control of the endogenous promoter is that it gives a visual expression profile for the gene of interest, enabling one to more easily decide on a suitable time point at which to begin stress treatment or determine the specific cell type within tissues where the target genes function. Even though we could not identify a defective phenotype in the *Os05g38530* mutant, we analyzed in planta expression patterns of the *Os05g38530*

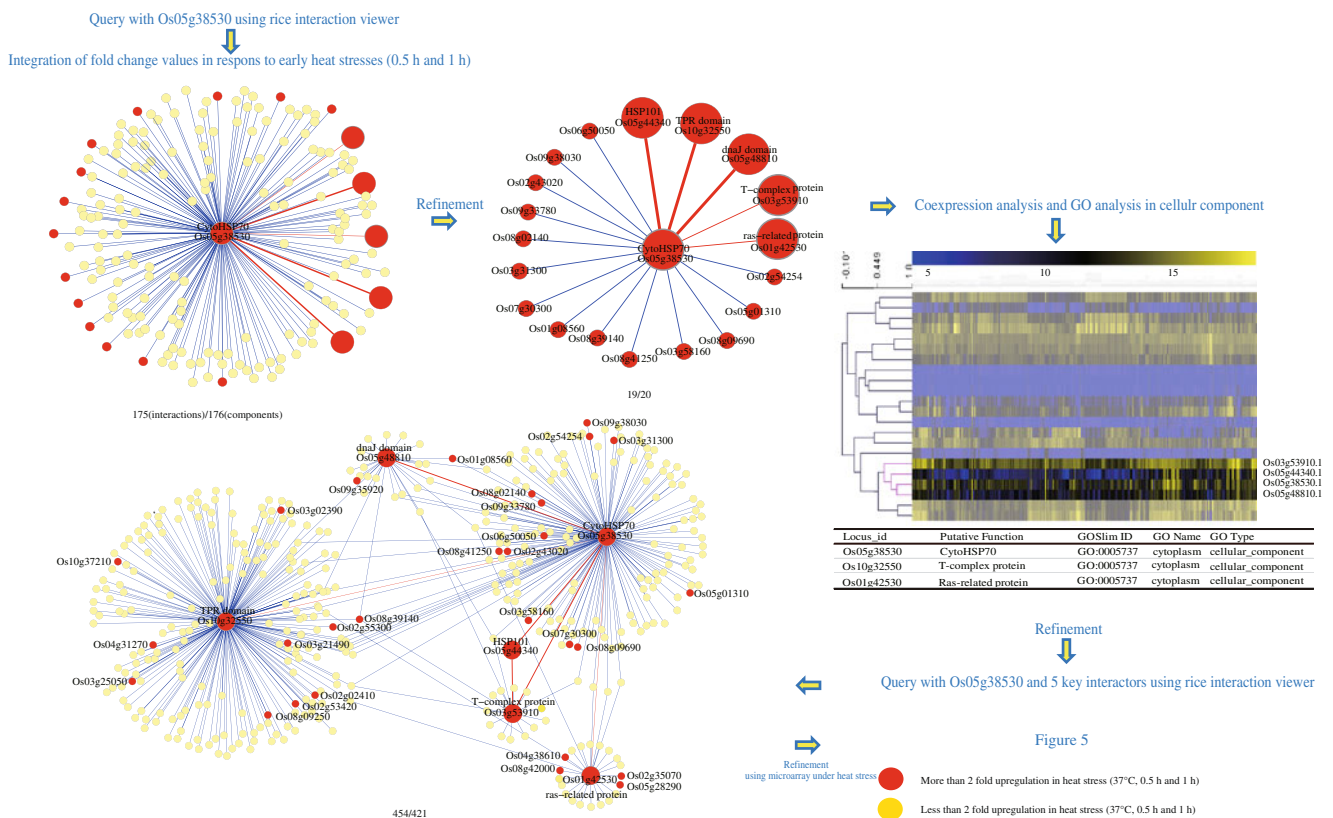


Fig. 4 Developing and refining the interaction network mediated by *OscHsp70-1*. Rice interaction viewer was primarily used to develop an interaction network mediated by *OscHsp70-1* (*Os05g38530*), for which 175 interactors were found. The network was refined by identification of 19 heat-inducible genes after integrating fold-changes in response to early heat stresses. Hierarchical clustering analysis with anatomical meta-profile for genes in the refined network revealed three with expression patterns most similar to *OscHsp70-1*, as well as two interactors predicted to be located in the cytoplasm, also similar to

OscHsp70-1. The network obtained by identifying proteins with five key interactors of *OscHsp70-1* was further expanded and refined by integrating fold-change data in response to heat, and by selecting proteins connected to at least two network proteins. Significant components in the expanded network are represented, with red node colors (34 genes) indicating at least 2-fold up-regulation in response to early heat stress (NSF45K array). Yellow nodes indicate less significant fold-changes. *HSP* heat shock protein, *TPR* tetratricopeptide repeat

with HSP70 and c-Raf inhibits a critical intermediate in Ras-induced MEK signaling and plays an important role in the function of this tumor suppressor (Sato et al. 2011a). Bag1 (Bcl-2-associated athanogene-1), a co-chaperone for HSP70, coordinates signals for cell growth in response to stress by down-regulating the activity of Raf-1 (Townsend et al. 2004). This also indicates that the interaction of *OscHSP70-1* or Ras-related protein with Bag and Raf kinase modulates stress-related cell growth, including programmed cell death. Finally, interaction of the activation and DNA-binding domains of HSF1 with HSP70 in *Arabidopsis* has been confirmed by sub-domain mapping after electrophoretic mobility shift and yeast two-hybrid assays (Kim and Schoffl 2002), suggesting that a heat stress transcription factor, *Os03g58160*, regulates the function of *OscHSP70-1*. In summary, a functional association between Ras/Raf-mediated MAPK kinase cascades and the HSP70 protein is suggested during transmission of a signal from a receptor on the cell surface to the DNA in the nucleus.

Among the components with multiple interactions, adenylate kinase (*Os07g22950*) is a phosphotransferase enzyme that catalyzes the interconversion of adenine nucleotides and also plays an important role in cellular energy homeostasis recognized by the ATP-dependent molecular chaperone (Papp et al. 2003). In that process, interaction between the Ras-related protein and the AMP-binding enzyme (*Os02g32490*) might be an upstream event that provides adenylate kinase with ADP, finally transferring ATP to *OscHSP70-1*. AGC kinase (*Os07g48290*) may be one of the modulators that phosphorylates *OscHSP70-1* with association of a T-complex protein. The interaction of SKP1-like protein 1b or the ubiquitin family domain-containing protein with *OscHSP70-1* suggests a role for these proteins in the ubiquitination process, as predicted by identification of co-chaperones of HSP90/HSP70 in a recent in silico analysis (Prasad et al. 2010). The interaction with the importin subunit beta protein (*Os05g28510*) suggests that *OscHSP70-1* is possibly translocated with its associates from the cytoplasm to the nucleus. This is similar to the role of *Arabidopsis* stromal HSP70 in the translocation of chloroplast proteins (Su and Li 2010). Finally, the interaction of *OscHSP70-1* with several hydrolase or dehydrogenase proteins might activate tolerance responses against various stresses, including heat. In silico prediction of the *OscHSP70-1*-mediated interaction network implies that the HSP70 chaperone machinery has a diverse range of functions.

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